

CITYTWIN · DESIGN PRECIS

CityTwin design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
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Source	docs/02-concept.md @ 9033288		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, privacy rules, kiosk massing model, first-order numbers, safety, open questions)
0.1	2026-09-25	Amish Chadha	Initial scaffold

CityTwin design precis

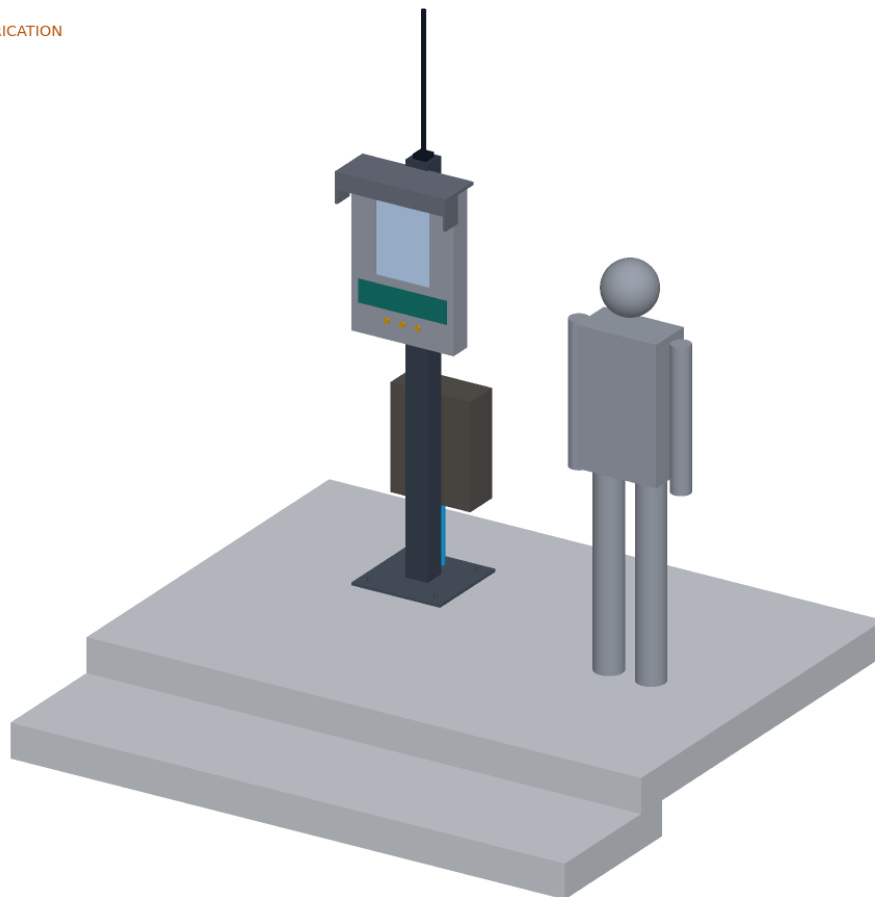
■ Summary

CityTwin is open software that runs on a TwinKit gateway and brings the lab's smart city nodes onto one map, plus a public street kiosk that shows the same data to residents. Nodes send counts and levels over LoRaWAN; CityTwin checks every record against a schema, stores it, publishes hourly open data files and draws the map. The kiosk is a steel post with a 13.3 in color e-paper screen at eye height, three push buttons, a notice plate that says what is measured and by whom, and a locked cabinet that holds the mains protection and the gateway.

First-order estimates, to be checked at TRL 3: a 46-node reference neighborhood sends about 4,900 records a day, well within one gateway; the kiosk and gateway draw about 9 W from the mains; the kiosk parts cost about \$740, or about \$1,025 with the TwinKit gateway, above the \$600 budget. The color e-paper takes about 20 s to redraw and is rated only 0 to 40 °C, so button response and the climate range are not met.

CityTwin

CONCEPT, NOT FOR FABRICATION



Grey: Person, 1.75 m (scale), sidewalk and curb for scale

Figure 1. CityTwin kiosk on a sidewalk, with a 1.75 m person for scale. Massing model.

■ How it works

1. **Collect.** The seven LoRaWAN node types (CurbCount, CrossSafe, LoadZone, HeatMap Node, FloodGauge, AirStreet, NoiseMap) send small uplinks to the TwinKit gateway's concentrator. PotholeLog uploads daily road summaries over depot Wi-Fi, and DockHub reports over LTE-M; both need an upload path agreed with those projects (open question).
2. **Check.** A decoder for each node type turns the payload into named fields. Any field outside that type's schema is rejected and logged, so a changed or faulty node cannot push anything but counts and levels into the store.
3. **Store.** Records go into the TwinKit time-series database with node ID, location and time. Raw records are kept for a limited period (proposed two years, awaiting Amish); hourly aggregates are kept indefinitely.
4. **Apply the privacy rules.** Before anything is published, people and vehicle counts below a threshold (proposed 5 per hour) become "fewer than 5", PotholeLog data is published only per road segment and per day, and DockHub data only per dock per hour. No card IDs, vehicle tracks or node-level raw streams leave the gateway.
5. **Publish.** Every hour the gateway writes CSV and GeoJSON files with a documented schema and pushes them one way to a public host under an open license. An OGC SensorThings API endpoint on the public host is proposed for TRL 3.
6. **Show.** The web map has one layer per node type, a time slider, node health and an "as of" timestamp. The kiosk controller pulls a pre-rendered 1600 x 1200 image of the selected layer from the gateway every 10 min, or when a button is pressed, and writes it to the e-paper. Each image carries its timestamp, so a stale screen during an outage says so.

CityTwin: data flow, records per day (estimates, 46-node reference neighborhood)

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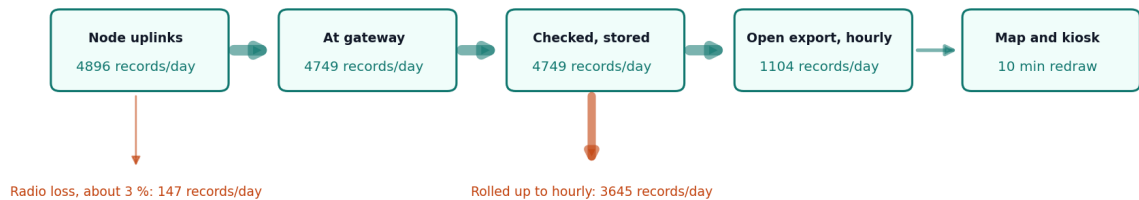


Figure 2. Data flow for the reference neighborhood, in records per day. All values are estimates: about 3 % radio loss, and about 4,750 stored records rolled up to about 1,100 hourly records for the open export.

■ Main components

Table 1. Main components. Numbers match `bom/bom.csv` and Figure 3.

#	Component	Proposed choice	Notes
1	Base plate and anchors	400 x 400 x 12 mm galvanized plate, four M16 anchors	Into a footing or existing slab
2	Post	100 x 100 x 4 mm square hollow section, 1.75 m	Carries head, hood, cabinet and antenna
3	Display head enclosure	Folded 2 mm aluminum, about 460 x 100 x 680 mm	IP54 target, vents underneath
4	Front window	6 mm UV-stabilized polycarbonate, anti-glare	Replaceable if scratched
5	E-paper display	13.3 in E Ink Spectra 6, 1600 x 1200 px, about 150 px per inch	Waveshare 13.3inch e-Paper HAT+ (E) : about 19 s full refresh, no partial refresh, 0 to 40 °C (manual)
6	Kiosk controller	Raspberry Pi Zero 2 W class, wired Ethernet to the gateway	Only pulls and draws images
7	Push buttons	Three 19 mm stainless buttons with LED rings, at about 1.12 m	Layer, time range, "about this sensor"
8	Data notice plate	400 x 100 mm, icons in the style of the open DTPR standard, QR code and short URL	Says what is measured, who is responsible and for how long data is kept
9	Sun and rain hood with light	Folded aluminum, 2 W LED strip with dusk sensor	Shades the screen by day, lights it at night
10	Services cabinet	Lockable steel, IP55, DIN rail, door on the back	Only place with mains voltage
11	Mains protection	6 A 30 mA RCBO and Type 2 surge protector	Installed or checked by an electrician
12	12 V DIN power supply	60 W, certified, SELV output	Feeds gateway and head
13	TwinKit gateway	TwinKit DIN gateway with LoRaWAN concentrator and LiFePO4 backup	Separate BOM (TwinKit); runs CityTwin software
14	LoRaWAN antenna	TwinKit antenna on a post-top bracket, top at about 2.4 m	Low for a gateway; see open questions
15	Conduit and cabling	Mains conduit into the cabinet; 12 V and Ethernet up the post	
16 to 18	Software and templates	Map dashboard, open data export, governance templates	MIT and CC licensed; no parts cost

CityTwin: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Base plate and anchors
- 2 Post, 100 mm square steel
- 3 Display head enclosure
- 4 Front window, polycarbonate
- 5 E-paper display, 13.3 in color
- 6 Kiosk controller
- 7 Push buttons (3)
- 8 Data notice plate
- 9 Sun and rain hood with light
- 10 Services cabinet, lockable
- 11 Mains protection (RCBO, surge)
- 12 12 V DIN power supply
- 13 TwinKit gateway with backup
- 14 LoRaWAN antenna (TwinKit)
- 15 Conduit and cabling

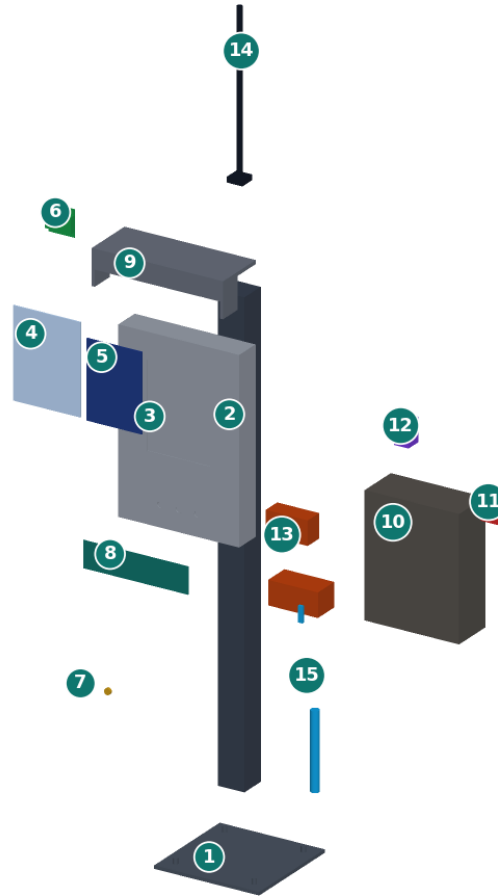


Figure 3. Exploded view with BOM numbers.

CityTwin: cutaway (section on the kiosk centerline, looking from the right)

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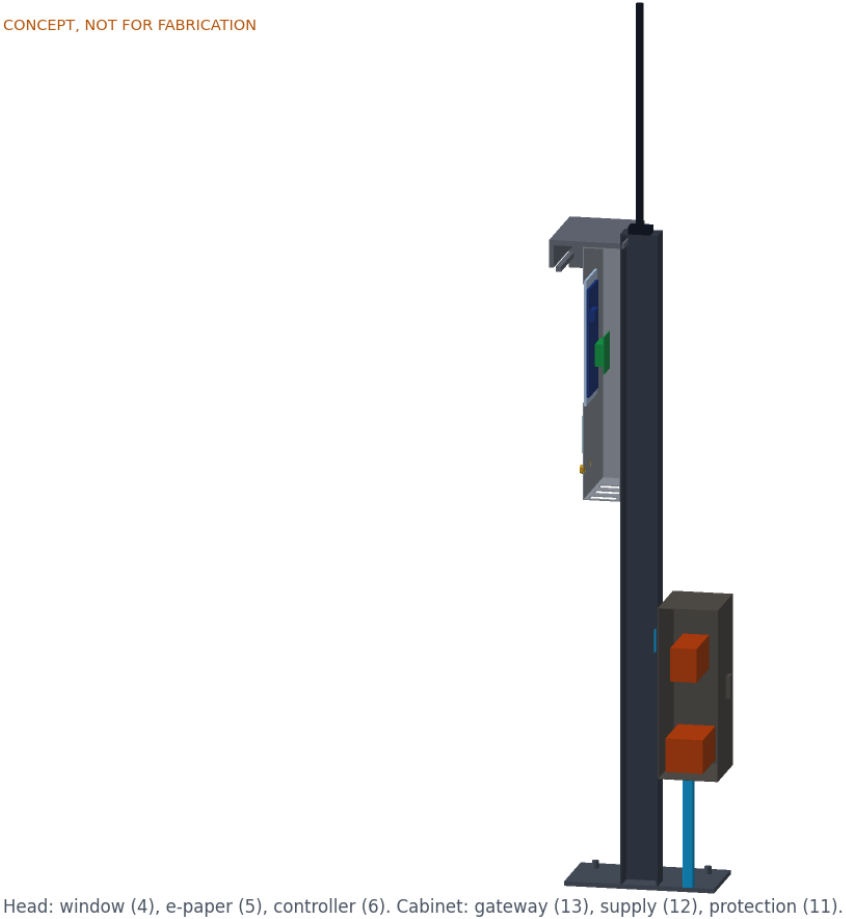


Figure 4. Section on the kiosk centerline, looking from the right: window (4), e-paper (5) and controller (6) in the head; gateway with backup pack (13), power supply (12) and mains protection (11) in the cabinet.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3. Assumptions are stated in each row and in CTW-REQ-001.

Table 2. Data load for the reference neighborhood (proposed).

Node type	Nodes	Records per node per day	Records per day	Basis
CurbCount	8	96	768	15 min counts (CurbCount README)
CrossSafe	4	24	96	Hourly activation summary (assumption)
LoadZone	12	64	768	State changes and heartbeats (assumption)
HeatMap Node	6	96	576	15 min means (HeatMap Node README)
FloodGauge	4	96	384	15 min status plus alerts (assumption)
AirStreet	6	288	1,728	5 min records (AirStreet README)
NoiseMap	6	96	576	15 min records (NoiseMap README)
Total	46		4,896	Within TwinKit's roughly 50 nodes (R2 met)

Table 3. System and kiosk estimates.

Quantity	Estimate	Basis	Requirement
Radio airtime	about 900 s per day, about 1 % of one channel, about 0.13 % over 8 channels	4,896 uplinks of about 20 bytes at SF9, about 185 ms each	R2 met with wide margin
Records stored	about 4,750 per day	3 % radio loss	
Storage	about 1 MB per day, about 350 MB per year	About 200 bytes per stored record with index	Fits the gateway's storage
Open export	about 1,100 hourly records per day, about 0.2 MB per day as CSV	46 nodes x 24 h	R6
Worst data age on screen	about 25 min	15 min node interval plus 10 min kiosk redraw	R8 met, thin margin
Button to new layer	about 20 s	About 1 s to fetch the image plus about 19 s full refresh	R10 not met (5 s)
Screen resolution	about 150 px per inch; 7 mm cap height is about 41 px	1,600 px over 270.4 mm	R9 by design
Kiosk power (head only)	about 1.8 W from the mains	Controller about 1.0 W, hood light 2 W for about 6 h a day (0.5 W average), button LEDs 0.1 W, e-paper refresh under 0.5 W for 19 s every 10 min (about 0.02 W), 88 % supply	
Kiosk and gateway	about 9 W, about 0.21 kWh per day, about 75 kWh per year	Adds TwinKit's about 6 W	R15 met
Wind on head and hood	about 350 N at about 1.4 m	0.35 m ² at 35 m/s gust, air 1.25 kg/m ³ , drag coefficient 1.3	
Post base moment and stress	about 0.58 kN·m; about 12 MPa	Adds about 160 N on the cabinet; section modulus of 100 x 100 x 4 SHS about 47,000 mm ³	R13 met by a wide margin
Anchor tension	about 1 kN per anchor	Moment over 0.3 m lever, two anchors in tension	Well within M16 anchors
Mass	about 55 kg	Post 21, base plate 15, cabinet 7, head 4.6, contents 4, hood 1.5, display and window 2 kg	Two-person lift
Kiosk parts cost	about \$740	Indicative, bom/bom.csv	R16 not met (\$600)
Kiosk and gateway	about \$1,025	Adds TwinKit's about \$285	R17 not met (\$600)

■ Key design choices

All are proposed, awaiting Amish.

- **Color e-paper rather than an LCD.** E-paper is readable in direct sun, draws almost nothing between refreshes and keeps its image in an outage. A 1,000-nit outdoor LCD would respond instantly and allow touch, but costs more, draws tens of watts and needs cooling. A monochrome e-paper panel would refresh faster but lose the color layers. Recommendation: color e-paper for the first kiosk, with an indoor LCD variant for libraries and city hall lobbies.
- **Buttons, not a touch screen.** Three stainless buttons survive vandalism and weather and suit e-paper's slow refresh; the QR code hands richer use over to the visitor's phone. Recommendation: buttons.
- **Gateway inside the kiosk cabinet.** One site, one mains feed and one locked box for a pilot. The cost is a low antenna (about 2.4 m), which shortens LoRaWAN range. The alternative is a rooftop gateway with the kiosk on a network link. Recommendation: gateway in the cabinet for the pilot, rooftop gateway for a full neighborhood.
- **Mains power.** Kiosk and gateway need about 0.21 kWh a day. A solar kiosk would need roughly a 75 W panel and a battery of about 0.6 kWh for three dark days (estimate), which does not fit the budget. The head alone (about 45 Wh a day) could run on a panel of about 30 W. Recommendation: mains, with a solar head-only variant for sites without power.
- **One-way publishing.** The gateway pushes open data files out and accepts no inbound connections from the internet, which keeps TwinKit's advice to keep the gateway off public networks until it is hardened. Recommendation: one-way push.
- **Privacy rules in the gateway, not only in the nodes.** Nodes already send counts and levels only; enforcing a schema and small-count suppression again at the gateway protects against a changed node or a future node type that is less careful. Recommendation: both layers.
- **Open data license.** Options: CC BY 4.0 (attribution, widely used for government data), CC0 (no conditions) or ODbL (share-alike). Recommendation: CC BY 4.0.
- **Budget.** See the review note for options.

■ Safety

Safety: The services cabinet contains mains voltage (230 V or 120 V). The mains feed, RCBO, surge protector and earthing must be installed or checked by a qualified electrician and follow local electrical code. Keep the cabinet locked, fit an earth bond to the post, head and hood, and carry only 12 V SELV up to the head.

Safety: The TwinKit gateway contains a LiFePO4 backup pack. Use a pack with a built-in BMS and fuse, charge it only through the TwinKit UPS module, and keep it within its temperature limits; a sealed cabinet in full sun can exceed them.

Safety: The kiosk is a roughly 55 kg steel post beside a walkway. Install it only with the asset owner's permission and a permit, on a footing designed for local wind and impact loads, with no sharp edges or protruding corners at head height, and keep a clear walkway width around it as local rules require. Lift it with two people or a hoist.

Safety: Privacy by design: no images, audio recordings or personal identifiers are taken in, stored or published; only aggregate counts or levels. Check local data protection law, and carry out a privacy impact assessment before any deployment. CityTwin is not a public warning system: do not rely on it for flood or emergency alerts.

■ Open questions

- [] Upload paths for PotholeLog (depot Wi-Fi files) and DockHub (LTE-M): agree formats with those projects rather than change them here.
- [] Reporting intervals and payload fields for CrossSafe, FloodGauge and LoadZone, which are assumed here.
- [] Climate range: a thermostatic heater and sun shading for the head, a panel rated for a wider range, or indoor siting only (R12).
- [] Button response: accept about 20 s with an LED acknowledgment, use a faster monochrome panel for part of the screen, or use an LCD (R10).
- [] Small-count threshold, raw data retention and the open data license. Proposed, awaiting Amish.
- [] Gateway in the kiosk cabinet or on a rooftop, and the antenna height needed for the reference neighborhood.
- [] Which languages and scripts the kiosk shows, and how residents ask questions or report a fault.
- [] First partner city or neighborhood, and co-design sessions with residents on what the kiosk should show.

Concept media: [blueprint sheet](#), [interactive 3D model](#).